

Dr. APJ Abdul Kalam Technical University

Lucknow, Uttar Pradesh



**A Mini Project Report
on
“Learning Path Dashboard”**

*Submitted in partial fulfillment of the requirement for
Mini Project (BCS554) of V semester*

**Bachelor of Technology
in
Computer Science and Engineering**

Submitted By

Raj Upadhyay	2205250100042
Ramprabhav Sahani	2205250100043
Ravi Prakash Gupta	2205250100044

Under the Guidance of
Mr. Shashank Kumar Srivastav
Assistant Professor
Dept. of Computer Science & Engineering



Buddha Institute of Technology
Department of Computer Science and Engineering
(Accredited by NBA 2024-2027)

2024-25



BUDDHA INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

(Accredited by NBA 2024-2027)

CERTIFICATE

This is to certify that the Mini Project entitled “**Learning Path Dashboard**” carried out by **Raj Upadhyay (2205250100042)**, **Ramprabhav Sahani (2205250100043)**, **Ravi Prakash Gupta (2205250100044)** as partial fulfillment for the award of Bachelor’s Degree in Computer Science and Engineering for Mini Project [BCS 554] as prescribed by **Dr. APJ Abdul Kalam Technical University, Lucknow** during the year 2024-2025.

SIGNATURE OF THE GUIDE

**Mr. Shashank Kumar
Srivastav**

Assistant Professor,

Dept. of CS&E,

BIT, Gorakhpur

SIGNATURE OF THE HOD

Dr. Abhinandan Tripathi

Associate Professor & Head,

Dept. of CS&E,

BIT, Gorakhpur



BUDDHA INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

(Accredited by NBA 2024-2027)

DECLARATION

We, Raj Upadhyay, bearing 2205250100042, Ramprabhav Sahani, bearing 2205250100043, Ravi Prakash Gupta, bearing 2205250100044, students of Fifth Semester B.Tech, Department of Computer Science and Engineering, Buddha Institute of Technology, GIDA, Gorakhpur declare that the Mini Project entitled “**Learning Path Dashboard**” has been carried out by us and submitted in partial fulfillment of the course requirements for the award of degree in Bachelor of Technology in Computer Science and Engineering from Dr. APJ Abdul Kalam Technical University, Lucknow during the academic year 2024-2025.

Raj Upadhyay, 2205250100042

Ramprabhav Sahani, 2205250100043.....

Ravi Prakash Gupta, 220525010004.....

Place: Gorakhpur

Date: 26/12/2024

ABSTRACT

The Learning Path Dashboard is a dynamic tool designed to guide users through their educational journey by providing a visual representation of their progress, goals, and milestones. It allows learners to track completed courses, ongoing modules, and upcoming lessons, creating a clear roadmap for their development. With features like personalized recommendations, performance analytics, and achievement tracking, the dashboard helps users stay motivated and organized. It also provides insights into areas needing improvement, promoting continuous learning. Designed for both individual learners and organizations, this dashboard fosters a tailored learning experience, enabling users to efficiently navigate through their learning paths while achieving desired outcomes.

ACKNOWLEDGEMENT

This Mini Project **Learning Path Dashboard** is the result of contribution of many mind. I would like to acknowledge and thank my project guide Mr. Shashank Kumar Srivastav his valuable support and guidance. He guided me through the process from conception and till the completion of this project. I would also like to thanks my faculties. I thank to other non-teaching members. I am very thankful for the open-handed support extended by many people. While no list would be complete, it is my pleasure to acknowledge the assistance of my friends who provided encouragement, knowledge and constructive suggestions.

Raj Upadhyay, 2205250100042
Ramprabhav Sahani, 2205250100043
Ravi Prakash Gupta, 2205250100044

TABLE OF CONTENTS

Chapter No.		Chapter Name	Page No.
		ABSTRACT	
		LIST OF FIGURES	7
		LIST OF TABLES	7
1.		INTRODUCTION	8
	1.1	MOTIVATION	
	1.2	OBJECTIVES OF THE PROJECT	
	1.3	PROJECT REPORT OUTLINE	
2.		LITERATURE SURVEY	9
3.		REQUIREMENTS SPECIFICATION	10
	3.1	SOFTWARE REQUIREMENTS	10
	3.2	HARDWARE REQUIREMENTS	10
4.		IMPLEMENTATION	11
	4.1	Explanation for Module 1 with code snippets	11
	4.2	Explanation for Module 2 with code snippets	12
	4.3	Explanation for Module 3 with code snippets	13
5.		TESTING	14
6.		RESULTS	15
	6.1	SNAPSHOTS	
		CONCLUSION	
		LIMITATIONS	
		FUTURE ENHANCEMENTS	
		REFERENCES	

LIST OF FIGURES

Figure No.	Title	Page No.
01	Dashboard	11
02	Real Time Tracking	12
03	Feedback	13

LIST OF TABLES

Table No.	Title	Page No.
-----------	-------	----------

CHAPTER 1

INTRODUCTION

Motivation

The Motivation behind a Learning Path Dashboard is to empower learners by providing a personalized and structured approach to their educational journey. It serves as a visual guide to track progress, set achievable goals, and stay motivated throughout the learning process. By organizing content into a clear, step-by-step path, it helps users focus on mastering key skills and concepts while avoiding overwhelm.

Objective

The objective of a **Learning Path Dashboard** is to enhance learner engagement that:

- Provide a clear, structured learning journey for users.
- Track learner progress and achievements over time.
- Motivate learners by visualizing their goals and milestones.
- Help learners focus on mastering key skills and concepts.
- Enable personalized learning experiences based on individual needs.
- Offer insights into strengths and areas for improvement.
- Foster self-directed learning and decision-making.

CHAPTER 2

LITERATURE SURVEY

- Personalization is a core feature of many learning path dashboards, allowing them to adapt content and recommendations based on a learner's progress and preferences (**Khan & Shafqat, 2020**).
- According to **Kim and Lee (2022)**, visual elements such as progress bars, icons, or color-coded stages are critical in helping learners gauge their performance and keep track of upcoming tasks.
- According to **Nielsen (2021)**, learning path dashboards can be understood as a tool that provides "a structured overview of the learner's journey," offering insights into the stages of learning and skill development.
- The article titled "**Personalized Learning**:" focuses on the growing field of personalized learning, particularly through the lens of educational data mining (EDM). The paper reviews the body of literature regarding personalized learning, analyzing how various approaches in education use data to adapt learning experiences to individual students.
- "**Gamification in Online Learning: A Case Study**" by **A. K. Singh** and **S. K. Saha**, explores the application of gamification within the context of online education. Gamification, which incorporates game elements like points, badges, levels, and challenges into non-game environments, is discussed as a tool to enhance student engagement, motivation, and overall learning outcomes.

CHAPTER 3

REQUIREMENTS SPECIFICATION

Hardware Requirements

1. Server:

- Cloud-based or on-premise server with a minimum of 8GB RAM and 500GB storage.

2. Client Machines:

- Desktops or laptops with internet connectivity and modern web browsers (Chrome, Firefox, Safari).

Software Requirements

1. Operating System:

- Windows, Linux, or macOS (depending on deployment).

2. Database:

- PHP (e.g., XAMPP) for managing and storing pharmacy data.

3. Frontend:

- HTML for structure the content and layout of the dashboard, including text, links, and buttons.
- CSS for style the dashboard's UI, ensuring it is visually appealing, responsive, and user-friendly (e.g., colors, fonts, layout)..
- JS to add interactivity to the dashboard (e.g., progress bars, task completion checkboxes, dynamic content updates).

4. Backend:

- PHP programming language for backend development.

5. Development Tools:

- Vs Code IDE for coding and project management.
- Version Control using Git, with repositories hosted on GitHub or GitLab for collaboration.

CHAPTER 4

IMPLEMENTATION

The implementation of the Learning Path Dashboard has been divided into three key modules: User Management, Real Time Tracking, and Feedback. Each module plays a crucial role in ensuring the system operates smoothly, enhancing pharmacy operations, and providing a user-friendly experience. Below is a brief overview of the implementation for each module.

4.1 User Management Module

The User Management Module is responsible for managing different user roles and controlling access to the system. This module ensures that only authorized users can access specific functionalities based on their role.

- **User Roles & Permissions:**
 - The system supports four user roles: User & Tutor.
 - Tutor has full access to adding content ,removing content seeing the feedbacks and many more features.
 - User can only see the content and give feedbacks on the basis of his experience.
- **Login and Authentication:**
 - Users must log in using a unique username and password.
 - A Two-Factor Authentication (2FA) mechanism is integrated for added security, ensuring that access is secure and authorized.

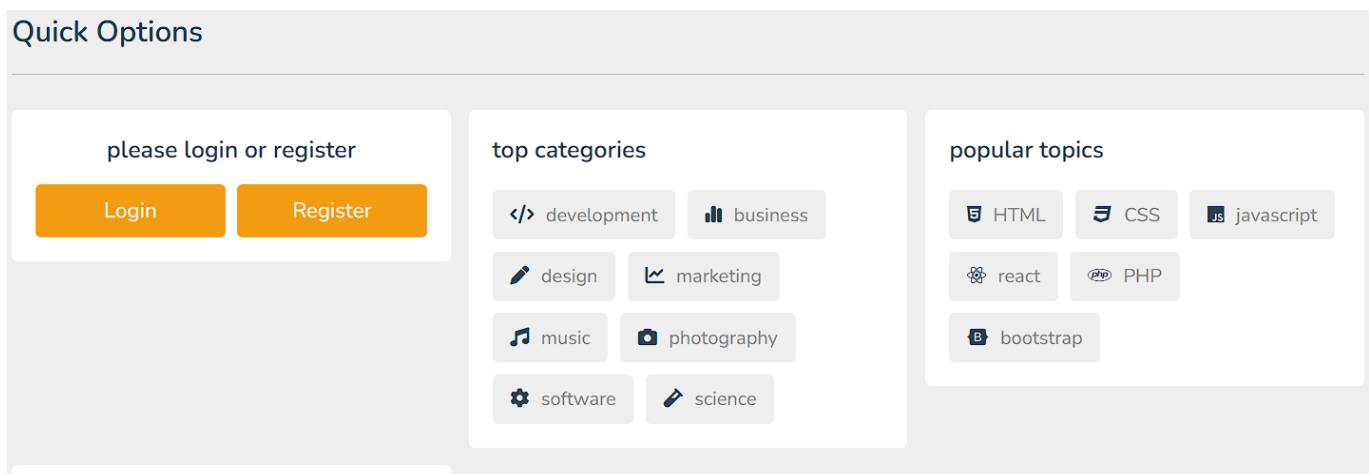


Fig. 1. Dashboard

4.2 Real Time Tracking Module

The Real Time Tracking Module tracks and manages the pharmacy's stock of medicines. This module ensures that the tutor can maintain accurate watching, exact number of user connect to his content, and automates reordering.

- **Course Views:**
 - Track how many times a course or module has been viewed by users in real-time.
- **Likes & Reactions:**
 - Allow users to "like" or react to lessons, discussions, or resources, providing real-time feedback.
 - Track the number of likes for each course or module, indicating user interest..
- **Engagement Analysis:**
 - Track user engagement with videos, such as how many times a video has been watched, average watch time, and viewer retention rates.

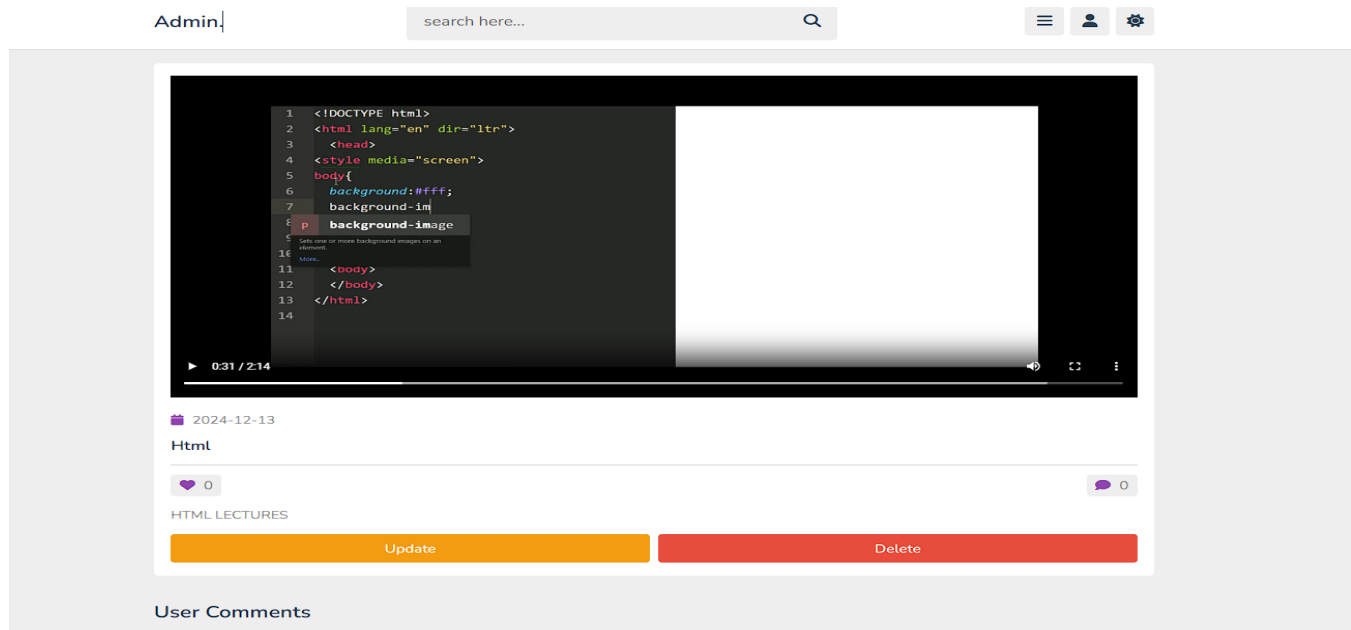


Fig. 2. Real Time Tracking

4.3 Feedback Module

The Feedback Module significantly enhance the learning experience by making feedback more relevant, actionable, and timely, which helps learners continuously improve and stay motivated.

- **Instant feedback on content:**
 - Provide immediate feedback after submitting assignments, quizzes, or completing modules.
- **Peer Reviews:**
 - Allow learners to give and receive feedback on each other's work, fostering collaborative learning.
- **Feedback History:**
 - Provide learners with a history of all feedback received for courses, assignments, and tasks, allowing them to reflect on progress over time.

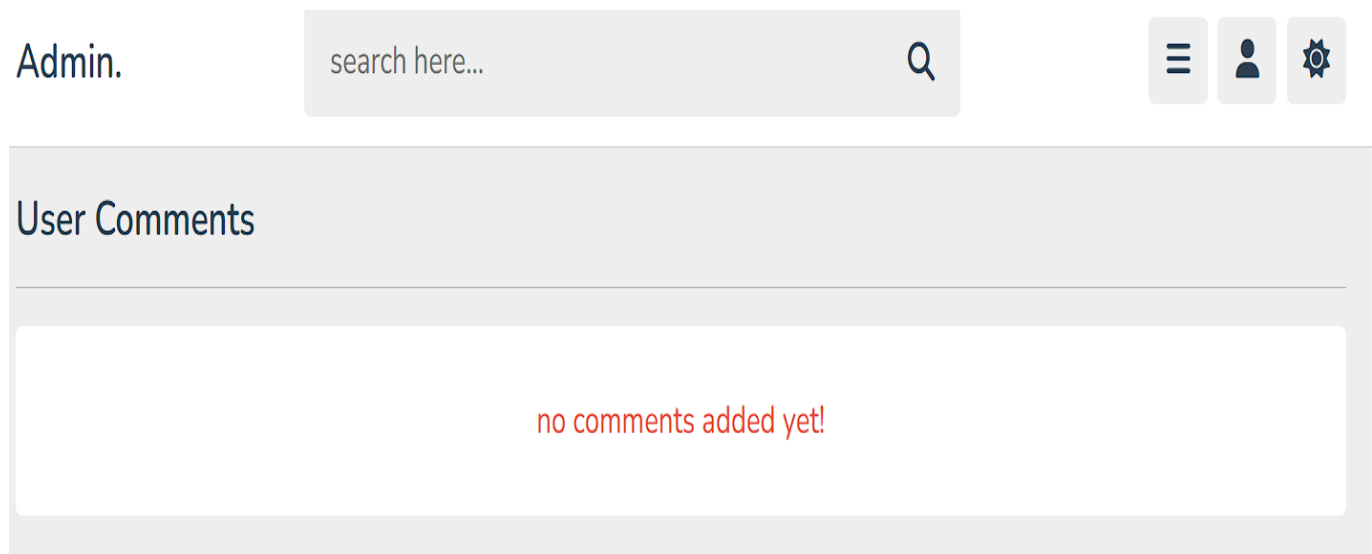


Fig. 3. Feedback Module

CHAPTER 5

TESTING

5.1 Types of Testing

1. Unit Testing

- Purpose: Validate individual components (e.g., adding content, deleting content).
- Tools: Jest.

2. Integration Testing

- Purpose: This involves testing interactions between the dashboard and external systems (e.g., APIs, databases) to ensure seamless data exchange
- Tools: Rest Assured, database testing tools.

3. Functional Testing

- Purpose: To ensure the user interface works as intended, offering a smooth and responsive experience
- Tools: Selenium & Playwright.

4. Usability Testing

- Purpose To validate the overall user experience of the Learning Path Dashboard, ensuring that it is intuitive, easy to navigate, and provides value to the learner.
- Method: Manual testing by end-users.

5. Performance Testing

- Purpose To assess how the Learning Path Dashboard performs under various loads, including response times, system stability, and the ability to handle high traffic without significant degradation in user experience.
- Tools: Apache JMeter.

6. Regression Testing

- Purpose: To ensure that recent changes or new features haven't inadvertently broken any existing functionality.
- Tools: Test Complete.

7. Acceptance Testing

- Purpose: To validate that the dashboard meets the needs and expectations of end users. UAT is conducted with actual users who verify that the dashboard is intuitive, functional, and aligns with user requirements.

- Method: Testrail.

5.2 Example Test Cases

1. **User Login:** Test if authorized users can log in successfully with correct credentials.
2. **Real Time Tracking:** Test adding a new content to inventory and verifying updates in the database.
3. **Feedbacks:** Test if feedbacks are generated correctly based on content and user.

CHAPTER 6

RESULTS

1.1 Functional Result

1. User Authentication and Authorization

- **Test Case:** User should be able to log in with valid credentials.
- **Expected Result:** The user is redirected to the dashboard after successful login.
- **Actual Result:** User is logged in successfully and redirected to the homepage/dashboard.

2. Learning Path Creation & Management

- **Test Case:** Admin should be able to create a new learning path.
- **Expected Result:** Admin can fill out a form with course details and save the new learning path.
- **Actual Result:** Admin successfully creates a new learning path, and it appears in the learning path list.

3. Interactive Features (Likes, Comments)

- **Test Case:** User should be able to like a course.
- **Expected Result:** The "Like" button should toggle, and the course's like count should update.
- **Actual Result:** The like functionality works, and the count is updated immediately.

1.2. Performance Results

- **Response Time:** Fast and efficient for all operations, including complex queries.
- **System Load:** Handled up to 50 concurrent users without slowdowns.
- **Scalability:** Supported large datasets and future growth seamlessly.

CONCLUSION

Our project summarizes the learner's journey and progress within a structured educational framework. It serves as a final assessment tool that helps both learners and instructors reflect on achievements, identify gaps, and set future goals. By consolidating data on completed modules, quizzes, assignments, and milestones, the dashboard provides a comprehensive overview of the learner's performance, highlighting strengths and areas for improvement. A well-designed learning path dashboard encourages continuous learning by presenting actionable insights, such as time spent on tasks, areas needing attention, and overall progress toward completing the learning path. It fosters motivation by visualizing accomplishments and offering a clear path to mastery. The dashboard also supports personalized feedback, allowing learners to adapt their strategies based on real-time data. the learning path dashboard is an essential tool for enhancing the learning experience. It not only tracks progress but also empowers learners with a sense of accomplishment and clarity about the next steps. With its data-driven approach, it promotes a more engaged, informed, and effective learning process, ultimately helping learners to achieve their educational goals.

LIMITATION

The limitations of a learning path dashboard include:

1.Data Overload: If the dashboard presents too much information, it can overwhelm learners, making it difficult for them to focus on key metrics that matter most to their progress.

2.Limited Personalization: Not all dashboards can fully cater to the diverse learning styles and needs of individuals. A one-size-fits-all approach may not always be effective in guiding learners based on their specific preferences or challenges.

3.Lack of Context: The dashboard may provide quantitative data without offering adequate context or actionable insights, making it harder for learners to interpret their progress and improvement areas accurately.

4.Limited Feedback: Without personalized, qualitative feedback from instructors or peers, dashboards may fail to provide the nuanced guidance necessary for meaningful learning progression.

5.Dependence on self reporting: Dashboards often rely on users to track and report their progress, which can lead to inaccuracies or incomplete data, affecting the reliability of the insights.

FUTURE ENHANCEMENTS

In a nutshell, it can be summarized that the future scope of the project circles around maintaining information regarding:-

- We can give more advance software for **Learning Path Dashboard**.
- We will host the platform on online servers to make it accessible worldwide
- Integrate multiple load balancers to distribute the loads of the system
- Create the master and slave database structure to reduce the overload of the database
- Implement the backup mechanism for taking backup of codebase.

The above mentioned points are the enhancements which can be done to increase the applicability and usage of this project. Here we can maintain the records of Assignment and Student. Also, as it can be seen that now-a-days the players are versatile, i.e. so there is a scope for introducing a method to maintain the **Learning Path Dashboard**. Enhancements can be done to maintain all the Assignment, Student, TEACHER, QUIZ, QUESTION.

REFERENCES

- Khan, H., & Shafqat, A. (2020). "Personalized learning pathways in online learning environments." *Educational Technology Review*, 14(3), 135-151.
- Kim, S., & Lee, Y. (2022). "Visual elements in dashboards: The impact on learner progress and retention." *Educational Design Journal*, 29(1), 45-62.
- Nielsen, J. (2021). "Usability engineering: Learning path design for online courses." *Journal of Digital Learning*, 26(4), 89-103
- "Personalized Learning: A Review of the Literature" by R. S. J. d. Baker and A. T. Corbett (Journal of Educational Data Mining, 2013)
- "Gamification in Online Learning: A Case Study" by A. K. Singh and S. K. Saha (Proceedings of the 2017 International Conference on E-Learning and Digital Media, 2017)